

30 Gas phase kinetics

Why Are We Learning This?

In the previous lecture, we introduced the idea that reactions occur through molecular events rather than simply through the overall stoichiometric equations written on paper. We learned that many reactions proceed through reaction mechanisms consisting of multiple elementary steps. In this lecture, we look more closely at what actually happens when molecules collide in the gas phase.

Most collisions do not immediately produce products. Instead, collisions often create short-lived, highly energized species that may either react or lose their energy and return to stable molecules.

Understanding these molecular processes helps explain:

- Why reaction mechanisms occur.
- How active intermediates form.
- Why reaction orders may change with concentration.
- Why the pseudo-steady-state hypothesis (PSSH) is useful.
- How molecular behavior influences observable reactor kinetics.

Course Roadmap

Introduction to CRE



General Balance Equation



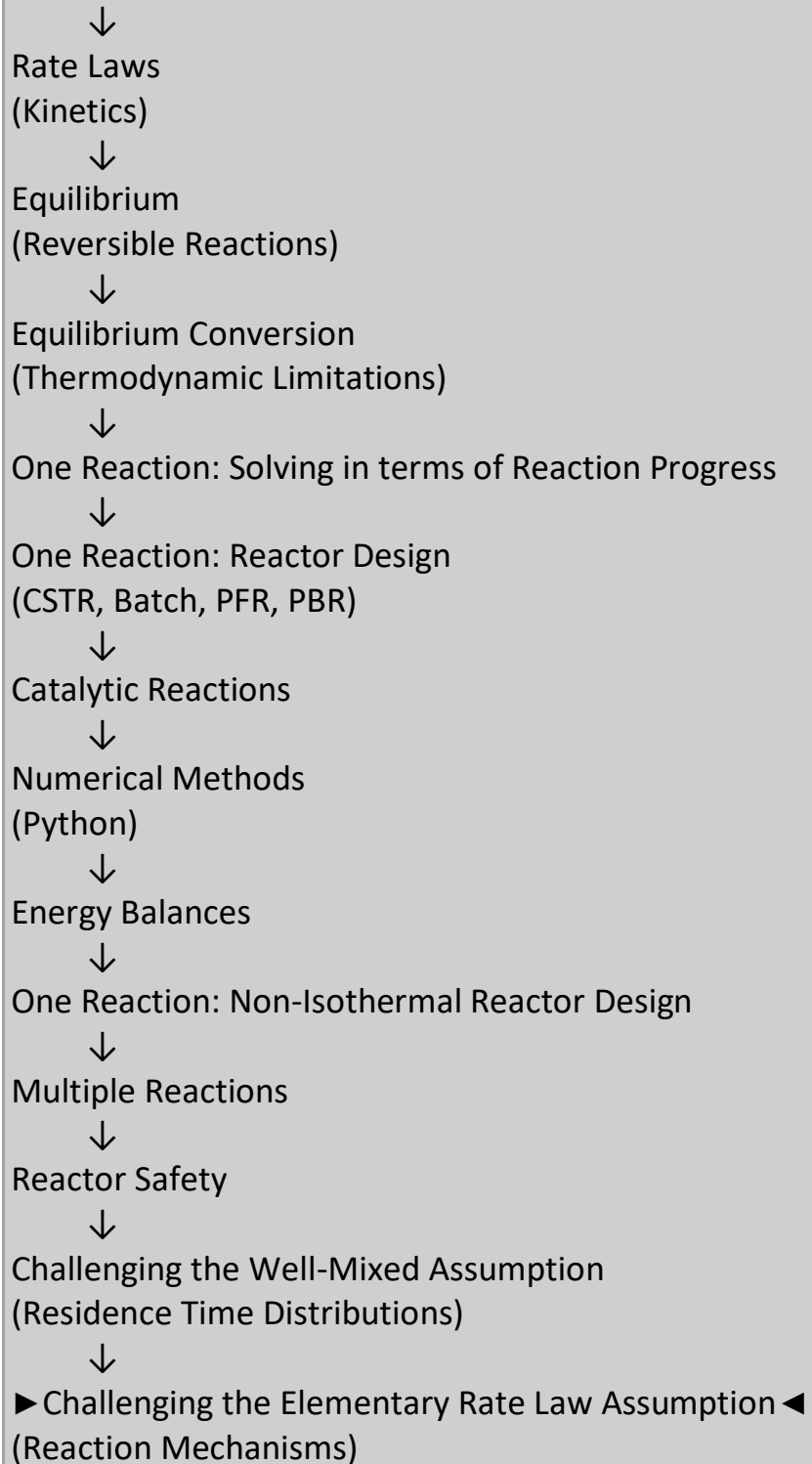
Common Reactor Models
(CSTR, Batch, PFR)



Reaction Progress
(Conversion)



Graphical Reactor Design
(Inverse Rate/Levenspiel Plots)



Learning Objectives

By the end of this lecture, you should be able to:

- Explain how molecular collisions lead to reactions.
- Define an active intermediate.
- Describe how collision energy is distributed within a molecule.
- Explain why energized species are often short-lived.
- Apply the pseudo-steady-state hypothesis (PSSH).
- Derive rate laws from gas-phase reaction mechanisms.
- Predict apparent reaction orders at high and low reactant concentrations.

Big Idea

Molecular Collisions



Active Intermediates



Reaction Mechanisms



Observed Rate Laws

Observable kinetics originate from molecular-level events occurring during collisions.

Collisions between gas species can result in a reaction, for example:

However, more often, collisions between gas molecules result in the formation of

Most Collisions Do Not Produce Products

It is tempting to imagine that every molecular collision results in a reaction. In reality: Most Collisions Do Not React

For a reaction to occur, molecules must have: Sufficient Energy, Correct Orientation, Appropriate Molecular Structure

Most collisions fail to satisfy all of these requirements simultaneously.

Physical Interpretation

Many Collisions

↓

Few Successful Collisions

↓

Chemical Reaction

This is why reaction rates are much slower than collision frequencies.

Key Insight

Collision Frequency $\neq$ Reaction Rate

Only a small fraction of molecular collisions result in successful chemical reactions because the molecules must collide with both sufficient energy and the proper orientation.

For example:

An active intermediate is

What Is an Active Intermediate?

An active intermediate is a highly energized molecular species formed during a collision. Rather than reacting immediately, collision energy is temporarily stored within the molecule.

Physical Interpretation

Collision

↓

Energy Transfer

↓

Excited Molecule

↓

Active Intermediate

These intermediates have very short lifetimes and either: React Further or Lose Their Excess Energy and return to stable molecules.

During the collision, kinetic energy is transferred into

where internal energy comprises

In other words, the energy is absorbed into

which causes

Where Does the Collision Energy Go?

When molecules collide, kinetic energy can be converted into internal molecular energy.

Internal energy includes: Bond Vibrations, Bond Rotations, Other Molecular Motions

The absorbed energy causes the molecule to become excited and temporarily unstable.

Physical Interpretation

Translational Motion

↓

Collision

↓

Vibrational Energy, Rotational Energy, Excited Electronic States

This energy storage is what produces active intermediates.

Active intermediates formed via collision can include

Due to their very short lifetimes, active intermediates can be modeled using the

The PSSH assumes

Why Is the PSSH Useful?

Active intermediates are often extremely short-lived. As a result: Formation Rate $\approx$ Consumption Rate for much of the reaction.

This observation leads to the: Pseudo-Steady-State Hypothesis (PSSH) which allows intermediate concentrations to be eliminated from rate-law derivations.

Key Insight

Intermediate Forms

↓

Intermediate Reacts Very Quickly

↓

Amount Present Remains Small

An intermediate species is

Let's do an example: The azomethane decomposition reaction is proposed to occur via the following mechanism:

1. $(\text{CH}_3)_2\text{N}_2 + (\text{CH}_3)_2\text{N}_2 \rightarrow (\text{CH}_3)_2\text{N}_2 + [(\text{CH}_3)_2\text{N}_2]^*$
2. $[(\text{CH}_3)_2\text{N}_2]^* + (\text{CH}_3)_2\text{N}_2 \rightarrow (\text{CH}_3)_2\text{N}_2 + (\text{CH}_3)_2\text{N}_2$
3. $[(\text{CH}_3)_2\text{N}_2]^* \rightarrow \text{C}_2\text{H}_6 + \text{N}_2$

What Does the * Mean?

The symbol * denotes an energized or activated species. For example:

$[(\text{CH}_3)_2\text{N}_2]^*$ is not a new chemical compound. Instead it represents: An Excited Form of Azomethane containing excess internal energy acquired during a collision.

Interpretation

Stable Molecule

↓

Collision

↓

Activated Molecule (*)

↓

Reaction or Deactivation

What is a mechanism?

What is happening in reaction 1? Why is $(\text{CH}_3)_2\text{N}_2$ both a reactant and a product??

Why Is the Same Molecule Both a Reactant and a Product?

At first glance, Reaction 1 appears strange because azomethane appears on both sides of the reaction. What is actually happening is:

Molecule 1 Transfers Energy To Molecule 2 during a collision. One molecule leaves unchanged, while the other becomes activated.

Physical Interpretation

Stable Molecule + Stable Molecule

↓

Collision

↓

Stable Molecule + Activated Molecule

The reaction represents energy transfer rather than chemical transformation.

What does the * mean?

What is happening in reaction 2?

What Is Deactivation?

Activated molecules do not always react. Sometimes they lose their excess energy through collisions with other molecules and return to their stable form.

Activated Molecule



Energy Loss



Stable Molecule

This process is called Deactivation and directly competes with product formation.

What is happening in reaction 3?

Product Formation Competes With Deactivation

The activated molecule has two possible fates:

Path 1: Deactivation: Return to a stable molecule.

Path 2: Reaction: Produce products.

Competition

Activated Molecule ↓	Activated Molecule ↓
Reaction ↓	Deactivation ↓
Products	Stable Molecule

The observed reaction rate depends on the balance between these competing pathways.

What are the intermediate species in the mechanism?

Write the rate law for the formation of ethane. Then, use the PSSH to eliminate the concentrations of all intermediate species from the rate law. What are the apparent reaction orders at high and low concentration of $(\text{CH}_3)_2\text{N}_2$?

Why Can Reaction Order Change?

One of the remarkable features of active-intermediate mechanisms is that the apparent reaction order can vary with concentration. At low concentration: Activation Events Are Rare

At high concentration: Collisions Become Frequent and the balance between activation, deactivation, and reaction changes.

Consequence

Same Mechanism



Different Apparent Reaction Orders depending on operating conditions.

Summary and Key Takeaways

Today we examined gas-phase reactions from a molecular perspective.

Key Ideas

- ✓ Most molecular collisions do not produce products.
- ✓ Collisions often produce energized active intermediates.
- ✓ Collision energy is stored as internal molecular energy.
- ✓ Active intermediates are typically short-lived.
- ✓ The pseudo-steady-state hypothesis can be used to eliminate intermediate concentrations.
- ✓ Activated species can either react or deactivate.
- ✓ Observed reaction orders may depend on concentration.
- ✓ Molecular mechanisms provide the link between molecular events and macroscopic rate laws.

Most Important Idea

Collisions

↓

Active Intermediates

↓

Reaction Mechanism

↓

Observed Kinetics

The reaction rate observed in a reactor is ultimately controlled by molecular events occurring during collisions, energy transfer, activation, deactivation, and product-forming reaction steps.